

Mental Health and Psychosocial Outcomes Among Individuals With Mayer-Rokitansky-Küster-Hauser (MRKH) Syndrome: A Scoping Review

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Background

MRKH syndrome is a rare Müllerian aplasia: congenital absence of the uterus and upper two-thirds of the vagina in individuals with a 46,XX karyotype and functional ovaries. It affects approximately 1 in 4,500–5,000 female births^{1,2} and is typically diagnosed in adolescence, a critical period for identity formation and social development.

The diagnosis carries profound implications for fertility, psychosexual development, and self-image, yet psychosocial outcomes remain underexplored and insufficiently integrated into clinical and public health frameworks.

MRKH Syndrome at a Glance

~1:5,000 female births	46,XX karyotype; functional ovaries	Teens typical age at diagnosis	No uterus uterovaginal agenesis
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Research Question: What mental health outcomes and coping mechanisms are reported among individuals with MRKH syndrome, and what healthcare system gaps exist?

Learning Objectives

- Identify key mental health outcomes, including depression, anxiety, reduced quality of life, and psychosexual challenges, reported among individuals with MRKH syndrome.
- Describe coping mechanisms and psychosocial responses, including adaptive and maladaptive strategies, documented in MRKH populations.

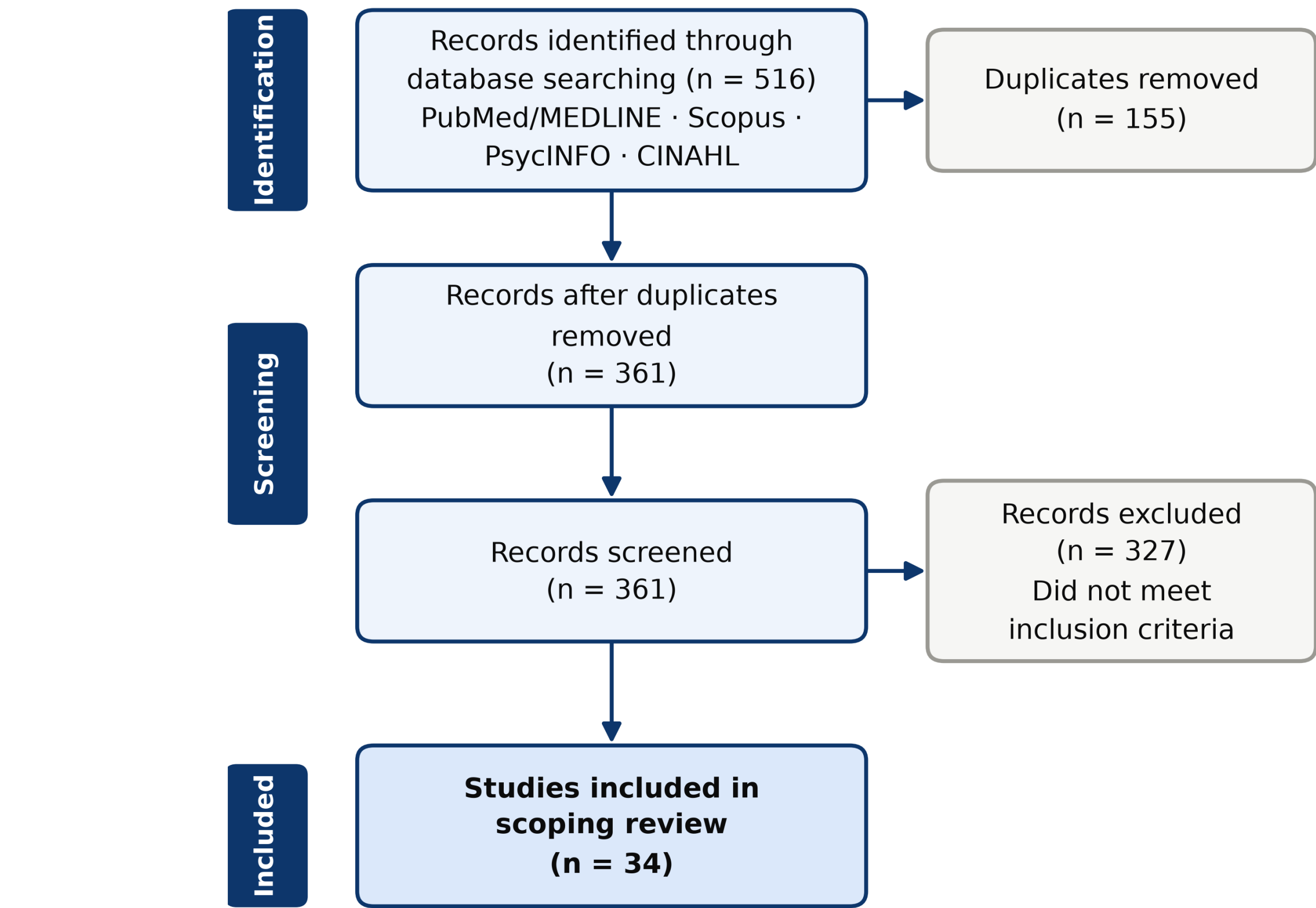
Methods

Design: Scoping review following Arksey & O'Malley's five-stage framework³; reported per PRISMA-ScR⁴.

Databases: PubMed/MEDLINE, Scopus, PsycINFO, CINAHL (January 2019 – March 2026).

Included: Primary quantitative, qualitative, and mixed-methods studies reporting psychological outcomes and/or coping mechanisms in MRKH populations.

Excluded: Studies focused solely on anatomical, surgical, or fertility outcomes.



Results

34 studies met inclusion criteria across quantitative, qualitative, and mixed-methods designs. Studies could report more than one outcome domain.

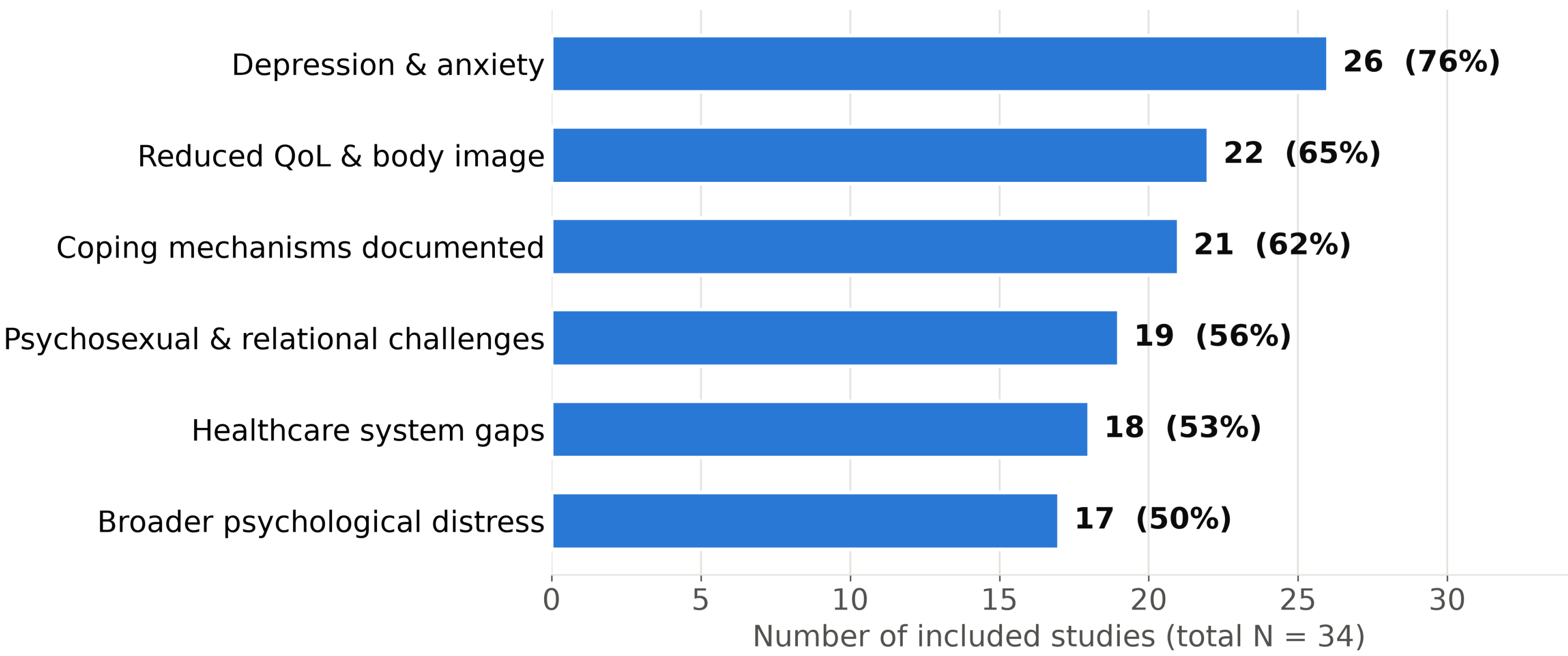


Fig 1. Frequency of outcome domains across included studies (N = 34). Counts do not sum to 34 because studies could report multiple domains.

Findings are consistent with pre-2019 evidence of elevated distress, anxiety, and depression and poorer mental health–related quality of life in MRKH^{5–7}, and with a prior systematic review⁸.

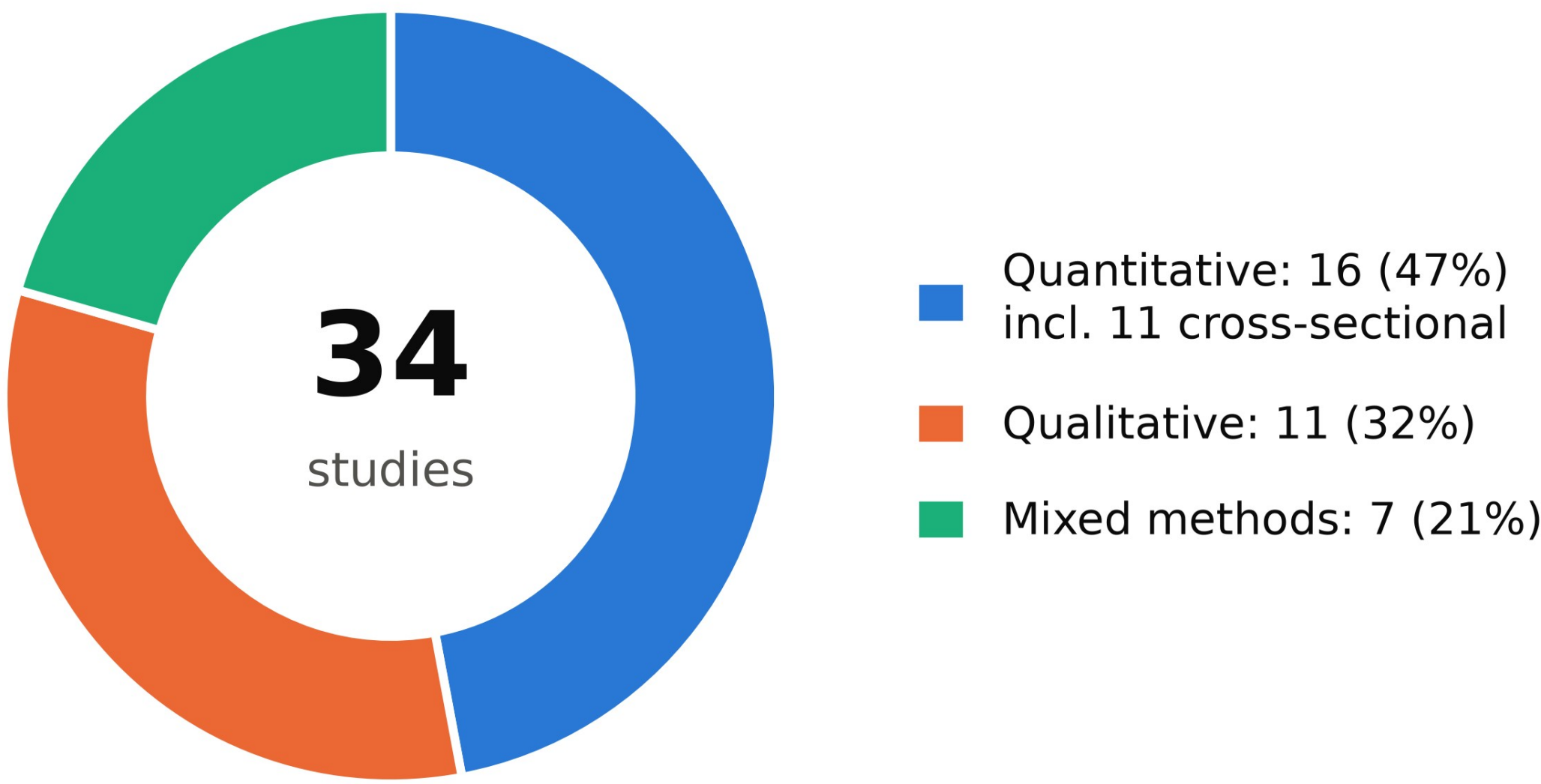


Fig 2. Study design distribution (N = 34). Cross-sectional studies are a subset of quantitative designs.

Coping Mechanisms (n = 21 studies, 62%)

Peer & community support	Psychological counseling
Identity reconstruction	Spiritual coping
Adaptive acceptance	Avoidance (maladaptive)

Blue = adaptive strategies; orange = maladaptive strategy.

Conclusions

MRKH-related psychosocial burden is substantial yet under-integrated into care models. Depression, anxiety, reduced quality of life, and psychosexual challenges are frequently reported and co-occurring, while documented coping mechanisms remain unsupported by systematic clinical pathways.

Multidisciplinary, mental health–inclusive care and greater research investment are urgently needed, particularly in underrepresented and low-resource settings.

Geographic Distribution of Studies

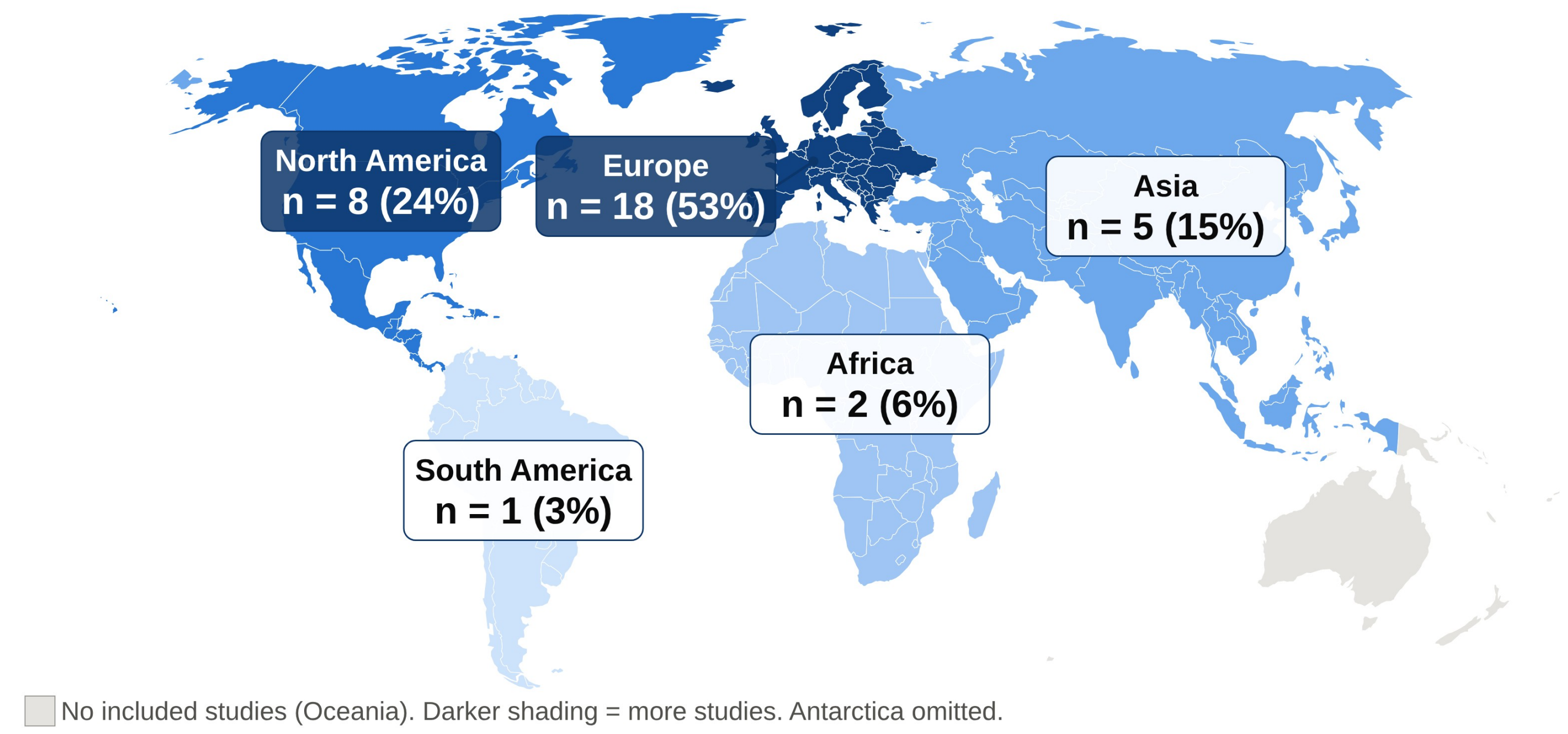


Fig 3. Included studies by region (N = 34). 26 of 34 (76%) originated in Europe or North America; only 3 (9%) came from Africa or South America.

Healthcare System Gaps (n = 18 studies, 53%)

- Delayed diagnosis in adolescence without concurrent psychosocial support
- Limited access to multidisciplinary care teams (psychology, social work, gynecology)
- Mental health insufficiently integrated into MRKH care models
- Lack of standardized mental health screening protocols at diagnosis

Public Health Implications

- Integrate mental health screening into standard MRKH diagnostic pathways
- Develop multidisciplinary care protocols including psychology and social work
- Expand research investment in low- and middle-income and underrepresented settings
- Advocate for peer support infrastructure within reproductive health services
- Train providers in psychosocially informed MRKH care

Limitations

- Review limited to 2019–2026; earlier foundational research not synthesized
- Heterogeneous designs and outcome measures limit direct comparison
- Most studies from high-income countries, limiting global generalizability
- Several studies relied on self-reported mental health outcomes

Key Takeaways

- About 3 in 4 studies (26/34) reported depression or anxiety
- Coping strategies were documented in 21/34 studies, yet remain unsupported by systematic clinical pathways
- Only 3 of 34 studies came from Africa or South America

References: ¹Herlin M, et al. Hum Reprod. 2016;31(10):2384–2390. ²ACOG Committee Opinion No. 728. Obstet Gynecol. 2018;131(1):e35–e42. ³Arksey H, O'Malley L. Int J Soc Res Methodol. 2005;8(1):19–32. ⁴Tricco AC, et al. Ann Intern Med. 2018;169(7):467–473. ⁵Heller-Boersma JG, et al. Psychosomatics. 2009;50(3):277–281. ⁶Laggari V, et al. J Psychosom Obstet Gynaecol. 2009;30(2):83–88. ⁷Liao LM, et al. Am J Obstet Gynecol. 2011;205(2):117.e1–6. ⁸Facchin F, et al. J Health Psychol. 2021;26(1):26–39.

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